

Levelling up

Everything the course assumes you already have: powers, expanding, factoring, fractions, equations, inequalities and systems.

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- 0.1 Powers and roots
 - 0.2 Working with algebraic expressions
 - 0.3 Factoring
 - 0.4 Fractions
 - 0.5 Linear equations
 - 0.6 Quadratic equations
 - 0.7 Turning a situation into an equation
 - 0.8 Inequalities and percentages
 - 0.9 Systems of linear equations
 - 0.10 Test yourself
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0.1 Powers and roots

LEARN IT ON KHAN ACADEMY

▶ **Video** The five rules · Exponent properties review

▶ **Video** Simplifying square roots

PRACTISE IT ON KHAN ACADEMY

 **Exercise** The rules together · Properties of exponents challenge (integer exponents)

 **Exercise** Simplify square roots

- 1 Same base, multiplied: add the exponents.

$$a^m \cdot a^n = a^{m+n}$$

- 2 Same base, divided: subtract them.

$$\frac{a^m}{a^n} = a^{m-n}$$

- 3 A power of a power: multiply them.

$$(a^m)^n = a^{mn}$$

- 4 A product inside a power: the power reaches every factor.

$$(ab)^n = a^n b^n$$

- 5 Zero and negative exponents are definitions, not tricks.

$$a^0 = 1 \quad (a \neq 0) \quad a^{-n} = \frac{1}{a^n}$$

- 6 A root is a fractional exponent, which is why the same five rules cover roots.

$$\sqrt[n]{a} = a^{1/n} \quad \text{the } n\text{-th root of } a = a^{1/n}$$

SIMPLIFY $\frac{(2x^3)^4}{4x^5}$

$$\frac{16x^{12}}{4x^5}$$

The fourth power reaches the 2 as well as the x^3 , so $2^4 = 16$ and x^{12} .

$$4x^7$$

Divide the numbers, subtract the exponents.

WATCH OUT

• $(a + b)^2$ is not $a^2 + b^2$. A power does not reach across a plus.

• a^{-n} is not negative. It is a reciprocal, so $2^{-3} = \frac{1}{8}$.

• $\sqrt{a^2 + b^2}$ does not split into $a + b$. Test it at $a = b = 3$.

0.2 Working with algebraic expressions

LEARN IT ON KHAN ACADEMY

► **Video** Expanding, from the start · Multiplying binomials intro

► **Video** Two brackets · Multiplying binomials

≡ **Article** Worked through · Warmup: Multiplying binomials

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✍ **Exercise** Expanding · Multiply binomials

✍ **Exercise** The difference of two squares · Multiply difference of squares

TERM

What the + and – signs separate. In $3x^2 - 5x + 2$ there are three terms.

FACTOR

What multiplies the whole of something. Factors cancel; terms never do.

LIKE TERMS

Same letter, same power. $3x$ and $5x$ are alike; $3x$ and $3x^2$ are not.

- 1 **Distribute across every term in the bracket.**

$$a(b + c) = ab + ac$$

- 2 **Two brackets: every term of the first against every term of the second.**

$$(a + b)(c + d) = ac + ad + bc + bd$$

- 3 **Two patterns are worth knowing by sight.**

$$(a + b)(a - b) = a^2 - b^2 \quad (a \pm b)^2 = a^2 \pm 2ab + b^2$$

- 4 A minus in front of a bracket reaches every term inside it.

$$-(x - 4) = -x + 4$$

- 5 Collect like terms last. Nothing else combines.

EXPAND $(2x - 5)(3x + 4)$

$$6x^2 + 8x - 15x - 20$$

Each term of the first bracket against each term of the second.

$$6x^2 - 7x - 20$$

Collect the two x terms.

WATCH OUT

• $2(x + 3)$ is $2x + 6$, not $2x + 3$. The 2 reaches both terms.

• $(q + 3)^2$ is $q^2 + 6q + 9$. The middle term is the one people drop.

• $3x$ and $3x^2$ do not combine. Different power, different term.

0.3 Factoring

LEARN IT ON KHAN ACADEMY

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- ▶ **Video** What factoring is · Introduction to factoring

 - ▶ **Video** Factoring a quadratic · Factoring quadratics intro

 - ▶ **Video** The difference of two squares · Factoring quadratics with difference of squares

 - ≡ **Article** The method, written out · Factoring simple quadratics review
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PRACTISE IT ON KHAN ACADEMY

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- ✍ **Exercise** Common factor first · GCF factoring introduction

 - ✍ **Exercise** Factoring a quadratic · Factoring quadratics intro
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- 1 Take out the common factor first, every time.

$$6x^2 + 9x = 3x(2x + 3)$$

- 2 For $x^2 + bx + c$, find two numbers that multiply to c and add to b .

$$x^2 - 13x + 36 = (x - 4)(x - 9)$$

- 3 Difference of two squares.

$$a^2 - b^2 = (a - b)(a + b)$$

- 4 Perfect square.

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

- 5 **When the leading coefficient is not 1, split the middle term and group.** Find two numbers multiplying to ac and adding to b .
- 6 **Check by expanding.** It costs ten seconds and catches every sign error.

FACTOR $3x^2 + 11x - 4$

$$ac = 3 \times (-4) = -12$$

Two numbers multiplying to -12 and adding to 11 : 12 and -1 .

$$3x^2 + 12x - x - 4$$

Split the middle term using those two numbers.

$$3x(x + 4) - 1(x + 4)$$

Group in pairs and take each common factor out.

$$(3x - 1)(x + 4)$$

The bracket $x + 4$ is now itself the common factor.

WATCH OUT

- $x^2 + 9$ does not factor over the real numbers. A sum of two squares never does.
- Missing the common factor first makes every later step harder and sometimes impossible.
- $4x^2 - 16$ is $4(x - 2)(x + 2)$. Completely means completely.

0.4 Fractions

LEARN IT ON KHAN ACADEMY

► **Video** Dividing fractions by fractions

PRACTISE IT ON KHAN ACADEMY

✍ **Exercise** Adding and subtracting · Add and subtract fractions

✍ **Exercise** Dividing fractions

- 1 **Multiply straight across. No preparation needed.**

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

- 2 **Divide by turning the second one over.**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

- 3 **Only adding and subtracting need a common denominator.** You can add halves to halves, not halves to thirds.

- 4 **Cancel factors, never terms.** A factor multiplies the whole numerator or the whole denominator.

- 5 **Keep the minus sign in front.**

$$\frac{-a}{b} = \frac{a}{-b} = -\frac{a}{b}$$

SIMPLIFY $\frac{x^2 - 25}{x + 5}$

$$\frac{(x - 5)(x + 5)}{x + 5}$$

Factor first. Only now is $x + 5$ a factor rather than a term.

$$x - 5$$

Valid for $x \neq -5$, since the original was undefined there.

WATCH OUT

❖ $\frac{x + 5}{x}$ is not 5. The x on top is a term, so it does not cancel.

❖ Test any cancellation with a number. At $x = 1$, $\frac{1 + 5}{1} = 6$, not 5.

❖ Adding numerators and denominators separately is the single most expensive habit on this list.

0.5 Linear equations

LEARN IT ON KHAN ACADEMY

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- ▶ **Video** Unknown on both sides · Linear equations with variables on both sides
 - ▶ **Video** With brackets · Linear equations with parentheses
 - ≡ **Article** The method, written out · Multi-step equations review
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PRACTISE IT ON KHAN ACADEMY

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- ✍ **Exercise** Unknown on both sides · Equations with variables on both sides
 - ✍ **Exercise** Unknown in a denominator · Rational equations intro
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- 1 **Whatever you do, do it to both sides.** That is the only rule; the rest is order.
- 2 **Clear brackets first.** $-(x - 4) = -x + 4$.
- 3 **Then clear fractions by multiplying every term by the common denominator.** Every term, including the ones on the right.
- 4 **Collect the unknown on one side and the numbers on the other.**
- 5 **Divide by the coefficient of the unknown.**
- 6 **If the unknown sits in a denominator, multiply through by it, then check.** A value that makes any original denominator zero is not a solution.

SOLVE $\frac{2x + 1}{3} - \frac{x - 4}{2} = 1$

$$2(2x + 1) - 3(x - 4) = 6$$

Multiply every term by 6, the common denominator.

$$4x + 2 - 3x + 12 = 6$$

Expand. The minus reaches both terms of the second bracket.

$$x + 14 = 6$$

Collect.

$$x = -8$$

Check: $\frac{-15}{3} - \frac{-12}{2} = -5 + 6 = 1.$

WATCH OUT

- Multiplying by the common denominator and forgetting the right-hand side.
- $-3(x - 4)$ is $-3x + 12$. Both signs change.
- With the unknown in a denominator, an answer that makes it zero must be thrown away.

0.6 Quadratic equations

LEARN IT ON KHAN ACADEMY

- ▶ **Video** Solving quadratics by factoring
 - ▶ **Video** The formula · The quadratic formula
 - ≡ **Article** The method, written out · Solving quadratics by factoring review
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PRACTISE IT ON KHAN ACADEMY

- ✍ **Exercise** Quadratics by factoring
 - ✍ **Exercise** Using the formula · Quadratic formula
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- 1 Get everything to one side.

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- 2 Read off a, b and c with their signs. The signs are where the marks go.
- 3 Try factoring. If it factors, it is faster. A product is zero only when one of its factors is zero.
- 4 Otherwise use the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- 5 The discriminant answers how many roots before you find them.

$$\Delta = b^2 - 4ac \quad \Delta > 0: \text{ two} \quad \Delta = 0: \text{ one} \quad \Delta < 0: \text{ none in } \mathbb{R}$$

SOLVE $2X^2 - 7X + 3 = 0$

$a = 2, \quad b = -7, \quad c = 3$

Read them off with their signs.

$\Delta = 49 - 24 = 25$

Positive, so there are two roots, and 25 is a square so they are tidy.

$x = \frac{7 \pm 5}{4}$

Put it in the formula.

$x = 3 \quad \text{or} \quad x = \frac{1}{2}$

WATCH OUT

- **$b = -7$ gives $b^2 = 49$, positive. Squaring kills the sign.**
- **$(x - 2)(x + 5) = 4$ tells you nothing. Only a product equal to *zero* splits.**
- **A negative Δ means no real solution, which in a business question usually means the target is never reached.**

0.7 Turning a situation into an equation

LEARN IT ON KHAN ACADEMY

► **Video** A situation into an equation · Linear equation word problems

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✍ **Exercise** Writing the equation · Writing linear equations word problems

✍ **Exercise** Writing and solving · Linear equations word problems

✍ **Exercise** Two-step problems · Two-step equations word problems

- 1 **Name the unknown, with its unit.** Write it down. "Let s = the number of soups sold."
- 2 **Write every other quantity in terms of it.** If 80 items sold and s are soups, the rest is $80 - s$.
- 3 **Find the sentence that says two things are equal. That is the equation.**
- 4 **Solve it.** Now it is only algebra.
- 5 **Answer in words, with the unit, and check it against the situation.** A negative count or a price of 0.003 means the setup was wrong, not the arithmetic.

A STALL SELLS ROLLS AT 4.50 AND SOUPS AT 7.00. IT SOLD 80 ITEMS AND TOOK 440.00. HOW MANY SOUPS?

Let s = soups, so rolls = $80 - s$

One unknown is enough; everything else follows from it.

$$7s + 4.5(80 - s) = 440$$

The total takings is the sentence that says two things are equal.

$$2.5s + 360 = 440$$

Expand and collect.

$$s = 32$$

32 soups and 48 rolls. Check: $32(7) + 48(4.5) = 224 + 216 = 440$.

WATCH OUT

- Starting to write algebra before defining the letter. Half the lost marks are here.
- Mixing units. If one price is in francs, all of them are.
- Answering with a bare number. The question asked how many soups.

0.8 Inequalities and percentages

LEARN IT ON KHAN ACADEMY

- ▶ **Video** Multi-step inequalities

 - ▶ **Video** Percentages · Percent problems

 - ≡ **Article** Double inequalities · Compound inequalities review
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PRACTISE IT ON KHAN ACADEMY

- ✍ **Exercise** Multi-step linear inequalities

 - ✍ **Exercise** Percentages · Percent word problems
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- 1 An inequality solves like an equation, with one exception.
- 2 Multiplying or dividing by a negative flips the sign.

$$-2x > 6 \quad \Rightarrow \quad x < -3$$

- 3 A double inequality: do the same thing to all three parts.

$$2 \leq 3x - 1 \leq 8 \quad \Rightarrow \quad 1 \leq x \leq 3$$

- 4 A percentage is a multiplier. That is the whole idea.

$$p\% \text{ of } A = \frac{p}{100} \cdot A$$

- 5 Up by $p\%$ multiplies; down by $p\%$ multiplies too.

$$\times \left(1 + \frac{p}{100}\right) \quad \times \left(1 - \frac{p}{100}\right)$$

- 6 Successive changes multiply. They never add.

A PRICE RISES 20%, THEN FALLS 20%. WHAT IS THE NET CHANGE?

×1.20 then ×0.80

Each change is a multiplier on what came before.

$$1.20 \times 0.80 = 0.96$$

Multiply the multipliers.

a 4% fall

Not zero. The fall is taken on the larger price.

WATCH OUT

- $-2x > 6$ gives $x < -3$. Dividing by the negative flips it.
- Up 20% then down 20% is not back where you started.
- A percentage point is not a percent. From 4% to 6% is two points and a 50% rise.

0.9 Systems of linear equations

LEARN IT ON KHAN ACADEMY

► **Video** By substitution · Solving systems of equations with substitution

PRACTISE IT ON KHAN ACADEMY

✍ **Exercise** By substitution · Systems of equations with substitution

✍ **Exercise** By elimination · Systems of equations with elimination

- 1 **Two unknowns need two independent equations.** Independent means neither is a multiple of the other.
- 2 **Substitution: make one equation say $x = \text{something}$, then put that into the other.** The other one.
Putting it back into the equation you rearranged gives $0 = 0$.
- 3 **Elimination: scale one or both so a letter cancels when you add or subtract.**
- 4 **If the letters vanish and a false statement is left, there is no solution.** $0 = 5$. The lines are parallel.
- 5 **If the letters vanish and a true statement is left, there are infinitely many.** $0 = 0$. It was the same line twice.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

SOLVE $3x + 2y = 19$ AND $x - y = 3$

$$x = y + 3$$

Rearrange the simpler of the two.

$$3(y + 3) + 2y = 19$$

Substitute into the *other* equation.

$$5y = 10, \quad y = 2$$

Expand and collect.

$$x = 5, \quad y = 2$$

Check both: $15 + 4 = 19$ and $5 - 2 = 3$.

WATCH OUT

- Substituting back into the equation you just rearranged. It always gives $0 = 0$.
- Subtracting equations and forgetting that the signs on the right change too.
- Checking in one equation only. A slip survives that.

0.10 Test yourself

Do this week's homework on the course platform, then check it against the answers when they are posted. That is the test. Everything above is how to be ready for it.



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